AIMLCZG512 – Deep Reinforcement Learning

Case Study - 1

Dynamic Programming for Chronic Disease Treatment Decisions

Type 2 Diabetes Cardiovascular-Risk Medication MDP

This case study formulates a repeated medication-initiation decision as a finite Markov Decision Process. The objective is to compute a state-dependent treatment policy using Bellman backups, backward Dynamic Programming, value iteration, or policy iteration.

Case idea

Patients with type 2 diabetes are reviewed periodically. At each review, the clinician observes metabolic risk factors and medication status, then decides whether to initiate cholesterol or blood-pressure medication. The patient's future health state evolves probabilistically. This naturally gives a finite Markov Decision Process, and Dynamic Programming can be used to compute an optimal treatment policy.

Topic: Dynamic Programming

Main reference: Steimle and Denton, *Markov Decision Processes for Screening and Treatment of Chronic Diseases*, 2017

Focus: Finite MDPs, Bellman backups, policy iteration, value iteration

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1. Topics Covered

1. How a chronic-disease treatment problem can be expressed as a sequential decision problem.
2. The MDP components: states, actions, transition probabilities, rewards, discounting, and horizon.
3. Bellman optimality equation for the treatment decision problem.
4. How value iteration or backward Dynamic Programming obtains an optimal policy.
5. How policy iteration can be used when the model is treated as stationary.
6. Interpretation of the resulting policy as a state-dependent clinical decision rule.

2. Problem Description

Patients with type 2 diabetes have increased risk of cardiovascular complications such as coronary heart disease (CHD), stroke, and premature death. In routine care, the physician observes the patient's current metabolic condition, including cholesterol and blood-pressure related factors, and then decides whether to initiate suitable medications.

This decision is not a one-time classification problem. It is a repeated decision problem. A medication started now may reduce future risk of CHD or stroke, but it may also create treatment cost, side effects, and medication burden. Delaying medication may be reasonable for a low-risk patient, but it may be harmful for a high-risk patient.

Sequential decision question

At each clinical epoch, given the patient's current metabolic state and medication status, which medication, if any, should be initiated so that the long-term expected health outcome is maximized?

The treatment process can be represented as shown in Figure 1. At epoch t, the patient state is observed, a medication decision is made, and the state evolves probabilistically before the next decision epoch.

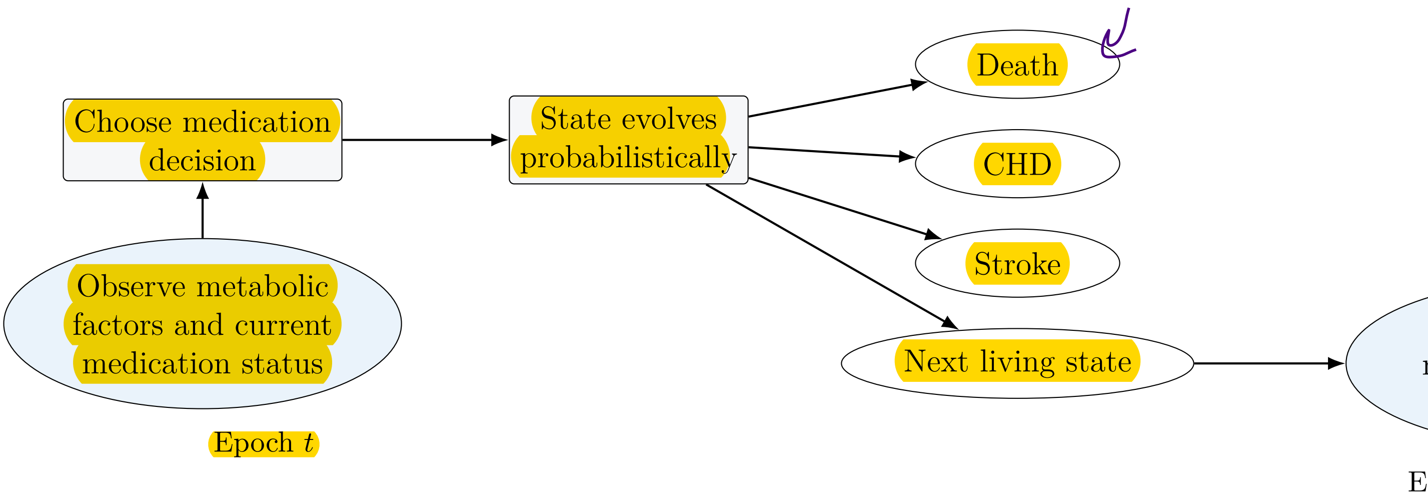

Figure 1: Representative treatment decision cycle for type 2 diabetes management. At each decision epoch, the clinician observes the current patient state, chooses a medication initiation decision, and the patient moves probabilistically to a future health state.

3. Why Dynamic Programming Fits This Case

Dynamic Programming (DP) is appropriate when a finite MDP model is available. In this case, the model specifies how patient states change under each treatment action and how rewards or costs are assigned to health outcomes.

The medical decision is not judged only by immediate cost. It is judged by immediate cost plus future health value. For example, starting medication may have a small immediate negative cost, but it may reduce the probability of future stroke or CHD. Hence the value of an action depends on both the current reward and the expected value of future states.

Bellman idea in this case

The physician is not simply asking: "Which action looks best now?" The physician is asking: "If I choose this action now, what future patient states may follow, and what is the long-term value of those states?"

4. Detailed MDP Formulation

The treatment problem can be modelled as a finite-horizon MDP:

$$(\mathcal{S}, \mathcal{A}, P, R, \gamma).$$

Here, $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $P(j \mid s, a)$ is the probability of moving to next state j from state s after action a, $R(s, a, j)$ is the reward, and γ is the discount factor.

4.1. Decision Epochs

The decision epochs are periodic clinical reviews, commonly modelled as annual decision points:

$$t = 0, 1, 2, \dots, N.$$

At each epoch, the clinician observes the current patient state and selects a feasible medication initiation action. A finite horizon is natural because the patient is followed over a lifetime or over a long planning period.

4.2. State Space

The state captures the patient's current metabolic condition and current medication status. A detailed state may be written as

$$s = (\mathrm{TC}, \mathrm{HDL}, \mathrm{SBP}, m_1, m_2, \dots, m_M),$$

where TC is total cholesterol category, HDL is HDL cholesterol category, SBP is systolic blood pressure category, and m_i indicates whether medication i has already been initiated.

For a compact explanatory model, the living states can be simplified as

$$s = (\text{risk level}, \text{cholesterol-medication status}, \text{BP-medication status}),$$

where

$$\text{risk level} \in \{\text{Low}, \text{Medium}, \text{High}\}.$$

The complete state space contains living states and absorbing event states:

$$\mathcal{S} = \mathcal{S}_L \cup \mathcal{S}_D.$$

Here, $\mathcal{S}_L$ is the set of living patient states and $\mathcal{S}_D$ contains event or terminal states such as CHD, stroke, and death.

Table 1: State components for the treatment MDP

Component	Meaning
Metabolic state	Discretized patient risk profile, such as cholesterol category, HDL category, and blood-pressure category.
Medication status	Records whether each medication has already been initiated. This prevents unrealistic repeated initiation of the same drug.
Living states	Patient is alive and has not entered an absorbing event state.
Absorbing states	CHD event, stroke event, or death. Once entered, the model remains there or stops further treatment decisions.

4.3. Action Space

The action is the medication initiation decision. In a simple two-medication-group version,

$$\mathcal{A}(s) = \{\text{Start none}, \text{Start cholesterol medication}, \text{Start BP medication}, \text{Start both}\}.$$

The feasible action set depends on the current medication status. For example, if cholesterol medication has already been initiated, then the action "Start cholesterol medication" is no longer feasible. Thus the action set is state-dependent.

Table 2: Illustrative state-dependent action set

Current medication status	**Feasible actions**
No medication started	Start none, start cholesterol medication, start BP medication, start both.
Cholesterol medication already started	Start none, start BP medication.
BP medication already started	Start none, start cholesterol medication.
Both medication groups already started	Start none.

4.4. Transition Probabilities

After an action is selected, the patient's state evolves probabilistically. The patient may remain alive and move to another living state, experience stroke, experience CHD, or die from other causes.

Let $p_t^S(s, a_t)$ be the probability of stroke, $p_t^C(s, a_t)$ be the probability of CHD, and p_t^O be the probability of death from other causes. A useful transition structure is

$$P_t(j \mid s, a_t) = \begin{cases} [1 - p_t^S(s, a_t) - p_t^C(s, a_t) - p_t^O]p(j \mid s), & s, j \in \mathcal{S}_L, \\ p_t^S(s, a_t), & j = d^S, \; s \in \mathcal{S}_L, \\ p_t^C(s, a_t), & j = d^C, \; s \in \mathcal{S}_L, \\ p_t^O, & j = d^O, \; s \in \mathcal{S}_L, \\ 1, & s = j, \; s \in \mathcal{S}_D, \\ 0, & \text{otherwise.} \end{cases}$$

The first case says that if no major event occurs, the patient continues in a living state according to the metabolic progression model. The next cases capture stroke, CHD, and death. Absorbing event states remain absorbing.

4.5. Reward Function

The reward function should represent the clinical objective. A common choice is based on quality-adjusted life years (QALYs), medication cost, treatment burden, and event penalties. A general reward form is

$$R_t(s, a, j) = \text{QALY}(s, j) - \lambda \cdot \text{MedicationCost}(a) - \text{EventPenalty}(j).$$

For a compact illustrative version, this can be simplified as follows:

- positive reward for surviving a year without a major event,
- small negative reward for medication cost or side effects,
- large negative reward for CHD or stroke,
- terminal negative reward or zero continuation value for death.

This reward design shows the trade-off clearly: the model may accept small immediate medication costs if the action reduces large future event losses.

4.6. Objective

The goal is to find an optimal policy $\pi_t^*(s)$ that maximizes expected discounted total reward:

$$V_t^*(s) = \max_{\pi} \mathbb{E}\left[\sum_{k=t}^{N} \gamma^{k-t} R_k(S_k, A_k, S_{k+1}) \mid S_t = s\right].$$

The discount factor γ controls the importance of future health outcomes. In chronic disease management, γ is usually chosen close to 1 because long-term complications are clinically important.

5. Solving the MDP Using Dynamic Programming

5.1. Backward Dynamic Programming for a Finite Horizon

For a finite horizon, the natural solution is backward induction. Initialize the terminal values:

$$V_N(s) = 0$$

or use a clinically estimated terminal value. Then compute values backward for $t = N-1, N-2, \ldots, 0$:

$$V_t(s) = \max_{a \in \mathcal{A}(s)} \sum_{j \in \mathcal{S}} P_t(j \mid s, a) \left[R_t(s, a, j) + \gamma V_{t+1}(j) \right].$$

The optimal action is

$$\pi_t^*(s) = \arg\max_{a \in \mathcal{A}(s)} \sum_{j \in \mathcal{S}} P_t(j \mid s, a) \left[R_t(s, a, j) + \gamma V_{t+1}(j) \right].$$

Backward Dynamic Programming Algorithm

1. Set terminal value $V_N(s)$ for all states.
2. For $t = N-1, N-2, \ldots, 0$:
   (a) For each state s, evaluate every feasible medication action $a \in \mathcal{A}(s)$.
   (b) Compute the expected immediate reward plus discounted future value.
   (c) Select the action with the largest value.
3. Store the selected action as $\pi_t^*(s)$.

The output is a time-dependent policy table. It tells which medication decision should be taken at each age, risk state, and medication status.

5.2. Value Iteration View

If the problem is represented as a stationary discounted MDP, value iteration repeatedly applies the Bellman optimality backup:

$$V_{k+1}(s) = \max_{a \in \mathcal{A}(s)} \sum_{j \in \mathcal{S}} P(j \mid s, a) \left[R(s, a, j) + \gamma V_k(j) \right].$$

The iteration is stopped when the largest change in values becomes small:

$$\Delta = \max_s |V_{k+1}(s) - V_k(s)| < \theta.$$

The greedy treatment policy is then extracted as

$$\pi(s) = \arg\max_{a \in \mathcal{A}(s)} \sum_{j \in \mathcal{S}} P(j \mid s, a) \left[R(s, a, j) + \gamma V(j) \right].$$

5.3. Policy Iteration View

Policy iteration alternates between evaluating the current treatment policy and improving it.

Policy Iteration for the Treatment MDP

Step 1: Initialize a treatment policy. For example, start with a simple guideline-like rule: no medication for low risk, one medication for medium risk, and both medications for high risk.
Step 2: Policy evaluation. For the current policy π, solve

$$V^{\pi}(s) = \sum_{j \in \mathcal{S}} P(j \mid s, \pi(s)) \left[R(s, \pi(s), j) + \gamma V^{\pi}(j) \right].$$

Step 3: Policy improvement. Improve the policy using one-step lookahead:

$$\pi_{\text{new}}(s) = \arg \max_{a \in \mathcal{A}(s)} \sum_{j \in \mathcal{S}} P(j \mid s, a) \left[R(s, a, j) + \gamma V^{\pi}(j) \right].$$

Step 4: Repeat. If the policy changes, evaluate the new policy again. If the policy is stable, stop.

Policy iteration is useful because it separates two ideas: first understand the value of a current treatment rule, then improve that rule by acting greedily with respect to its value function.

6. Small Illustrative Example

Consider a simplified MDP with three living states and one event state:

$$\mathcal{S}_L = \{\text{Low}, \text{Medium}, \text{High}\}, \qquad \mathcal{S}_D = \{\text{Event}\}.$$

The actions are

$$\mathcal{A} = \{\text{Wait}, \text{Start C}, \text{Start B}, \text{Start Both}\},$$

where C denotes cholesterol medication and B denotes blood-pressure medication.

A simple reward design is shown below.

Table 3: Illustrative reward structure for the simplified example

Outcome or action component	Reward contribution
Survive one year without major event	$+1$
Medication cost for Start C	-0.03
Medication cost for Start B	-0.03
Medication cost for Start Both	-0.06
CHD or stroke event	-5

For a high-risk patient, the action “Start Both” may have higher long-term value despite its immediate cost, because it may reduce the probability of future CHD or stroke. For a low-risk patient, “Wait” may be preferred because the medication burden may not be justified by a small event-risk reduction.

Thus, the optimal policy is not one fixed action for all patients. It is a mapping from patient state to action.

7. Expected Output of the DP Solution

The DP solution is an optimal policy table. A simplified version may look like Table 4.

Table 4: Illustrative optimal policy table

Patient state	Illustrative optimal action
Low risk, no medication	Wait
Medium risk, no medication	Start cholesterol medication
High risk, no medication	Start both medications
High risk, cholesterol medication already started	Start BP medication
Any absorbing event state	No further treatment initiation decision

In the real model, the table will be much larger because it may include many cholesterol categories, blood-pressure categories, medication combinations, age effects, and risk equations. However, the interpretation remains the same: Dynamic Programming chooses the action that gives the best expected long-term value from each state.

8. Dynamic Programming Interpretation

This case study connects directly to the DP algorithms in reinforcement learning.

- **Policy evaluation:** Estimate the long-term value of following a given clinical treatment rule.
- **Policy improvement:** Change the treatment rule by choosing actions that have better one-step lookahead value.
- **Policy iteration:** Repeatedly evaluate and improve until the treatment rule stabilizes.
- **Value iteration:** Directly apply the Bellman optimality backup and extract the greedy treatment rule.
- **Generalized policy iteration:** Evaluation and improvement interact, whether done fully, partially, or approximately.

The main message is simple: DP is not only a grid-world method. It can be used for serious sequential decision problems when a model of transition probabilities and rewards is available.

9. Limitations and Practical Cautions

The MDP solution is only as good as the clinical model. Important limitations are:

1. Transition probabilities must be estimated carefully from clinical data.
2. Continuous medical measurements must be discretized, which may lose information.
3. Medication adherence may be imperfect.
4. Side effects and patient preferences may vary across patients.

5. Clinical validation is necessary before using such a model as decision support.

Hence, this should be understood as a decision-support framework, not as an automatic replacement for physician judgment.

10. Points to Remember

- A chronic disease treatment problem can be formulated as an MDP when states are observed and transitions are modelled probabilistically.
- Actions correspond to medication initiation decisions.
- Rewards represent QALYs, survival benefits, treatment burden, costs, and event penalties.
- Dynamic Programming computes the long-term value of actions, not merely their immediate effect.
- The optimal policy is a state-dependent treatment rule.

11. Review Questions

1. Identify the state, action, transition probability, and reward components in the type 2 diabetes treatment MDP.
2. Why is a medication decision a sequential decision problem rather than a one-step classification problem?
3. Write the Bellman optimality equation for the finite-horizon treatment problem.
4. Explain why starting medication may have lower immediate reward but higher long-term value.
5. Compare value iteration and policy iteration for this treatment decision problem.

12. References

1. Steimle, L. N., and Denton, B. T. (2017). *Markov Decision Processes for Screening and Treatment of Chronic Diseases*. In *Markov Decision Processes in Practice*, Springer, pp. 189–222. Available at: `https://btdenton.engin.umich.edu/wp-content/uploads/sites/138/2015/10/Steimle-2015.pdf`
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3. Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*, 2nd edition. MIT Press. Chapter 4: Dynamic Programming.